Customer Shopping Behavior Analysis

Created by : Abdul Wahab

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- **Rows:** 3,900 - **Columns:** 18 - Key Features:
- **Customer demographics** (Age, Gender, Location, Subscription Status)
- **Purchase details** (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- **Shopping behavior** (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- **Missing Data:** 37 values in Review Rating column

3. Exploratory Data Analysis using R

We began with data preparation and cleaning in R:

- **Data Loading:** The dataset was imported using the `read_xlsx()` function from the `readxl` package and converted into a `tibble` for better readability and manipulation.

- **Initial Exploration:** Used `glimpse()` to check structure and `summary()` for summary statistics.

```
Rows: 3,900
Columns: 18
$ `Customer ID`            <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, …
$ Age                      <dbl> 55, 19, 50, 21, 45, 46, 63, 27, 26, 57, 53, 30, 61, 65, 64, 64, 25, 53, 52,…
$ Gender                   <chr> "Male", "Male", "Male", "Male", "Male", "Male", "Male", "Male", "Male", "Ma…
$ `Item Purchased`         <chr> "Blouse", "Sweater", "Jeans", "Sandals", "Blouse", "Sneakers", "Shirt", "Sh…
$ Category                 <chr> "Clothing", "Clothing", "Clothing", "Footwear", "Clothing", "Footwear", "Cl…
$ `Purchase Amount (USD)`  <dbl> 53, 64, 73, 90, 49, 20, 85, 34, 97, 31, 34, 68, 72, 51, 53, 81, 36, 38, 48,…
$ Location                 <chr> "Kentucky", "Maine", "Massachusetts", "Rhode Island", "Oregon", "Wyoming", …
$ Size                     <chr> "L", "L", "S", "M", "M", "M", "M", "L", "L", "M", "L", "S", "M", "M", "L", …
$ Color                    <chr> "Gray", "Maroon", "Maroon", "Maroon", "Turquoise", "White", "Gray", "Charco…
$ Season                   <chr> "Winter", "Winter", "Spring", "Spring", "Spring", "Summer", "Fall", "Winter…
$ `Review Rating`          <dbl> 3.1, 3.1, 3.1, 3.5, 2.7, 2.9, 3.2, 3.2, 2.6, 4.8, 4.1, 4.9, 4.5, 4.7, 4.7, …
$ `Subscription Status`    <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes"…
$ `Shipping Type`          <chr> "Express", "Express", "Free Shipping", "Next Day Air", "Free Shipping", "St…
$ `Discount Applied`       <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes"…
$ `Promo Code Used`        <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes"…
$ `Previous Purchases`     <dbl> 14, 2, 23, 49, 31, 14, 49, 19, 8, 4, 26, 10, 37, 31, 34, 8, 44, 36, 17, 46,…
$ `Payment Method`         <chr> "Venmo", "Cash", "Credit Card", "PayPal", "PayPal", "Venmo", "Cash", "Credi…
$ `Frequency of Purchases` <chr> "Fortnightly", "Fortnightly", "Weekly", "Weekly", "Annually", "Weekly", "Qu…
```

```
 Customer ID          Age           Gender          Item Purchased       Category
Min.   :   1.0   Min.   :18.00   Length:3900        Length:3900        Length:3900
1st Qu.: 975.8   1st Qu.:31.00   Class :character   Class :character   Class :character
Median :1950.5   Median :44.00   Mode  :character   Mode  :character   Mode  :character
Mean   :1950.5   Mean   :44.07
3rd Qu.:2925.2   3rd Qu.:57.00
Max.   :3900.0   Max.   :70.00

Purchase Amount (USD)   Location            Size              Color             Season
Min.   : 20.00        Length:3900        Length:3900        Length:3900        Length:3900
1st Qu.: 39.00        Class :character   Class :character   Class :character   Class :character
Median : 60.00        Mode  :character   Mode  :character   Mode  :character   Mode  :character
Mean   : 59.76
3rd Qu.: 81.00
Max.   :100.00

Review Rating  Subscription Status Shipping Type      Discount Applied   Promo Code Used
Min.   :2.50   Length:3900         Length:3900        Length:3900        Length:3900
1st Qu.:3.10   Class :character    Class :character   Class :character   Class :character
Median :3.80   Mode  :character    Mode  :character   Mode  :character   Mode  :character
Mean   :3.75
3rd Qu.:4.40
Max.   :5.00
NA's   :37
Previous Purchases Payment Method     Frequency of Purchases
Min.   : 1.00      Length:3900        Length:3900
1st Qu.:13.00      Class :character   Class :character
Median :25.00      Mode  :character   Mode  :character
Mean   :25.35
3rd Qu.:38.00
Max.   :50.00
```

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
55	Nevada	L	Brown	Summer	NA	Yes	Free Shipping
57	Idaho	S	Silver	Spring	NA	Yes	2-Day Shipping
100	Virginia	XL	Green	Spring	NA	Yes	2-Day Shipping
77	New Hampshire	S	Gold	Spring	NA	Yes	Express
92	Idaho	L	Gold	Fall	NA	No	Express
97	Oklahoma	L	Violet	Winter	NA	No	Next Day Air
20	Iowa	M	Orange	Spring	NA	No	Express
81	Wisconsin	M	Brown	Summer	NA	No	Express
86	Ohio	M	Gray	Spring	NA	No	Express
83	Oregon	M	Red	Winter	NA	No	Free Shipping
97	Arkansas	XL	Teal	Summer	NA	No	Free Shipping
100	Wisconsin	S	Red	Winter	NA	No	2-Day Shipping
64	Kentucky	M	Green	Winter	NA	No	Express
80	Illinois	L	Brown	Spring	NA	No	Standard
99	Virginia	M	Charcoal	Spring	NA	No	Express
61	Kentucky	M	Maroon	Summer	NA	No	Free Shipping

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

```
 [1] "customer_id"            "age"                    "gender"                 "item_purchased"
 [5] "category"               "purchase_amount"        "location"               "size"
 [9] "color"                  "season"                 "review_rating"          "subscription_status"
[13] "shipping_type"          "discount_applied"       "promo_code_used"        "previous_purchases"
[17] "payment_method"         "frequency_of_purchases"
```

- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.

frequency_of_purchases	age_group	purchases_frequency_days
Fortnightly	Middle-aged	14
Fortnightly	Young Adult	14
Weekly	Middle-aged	7
Weekly	Young Adult	7
Annually	Middle-aged	365
Weekly	Middle-aged	7
Quarterly	Senior	90
Weekly	Young Adult	7
Annually	Young Adult	365
Quarterly	Middle-aged	90

- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

- **Database Integration:** Connected R script to BigQuery and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

Row	gender	total_revenue
1	Male	157890
2	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

Row	customer_id	purchase_amount
1	106	96
2	319	67
3	1005	79
4	1170	94
5	1428	90
6	1006	94
7	1090	61
8	1530	63
9	1532	83
10	549	64
11	621	89
12	1305	63
13	1117	96
14	403	78
15	881	75
16	1326	73
17	177	81
18	195	94
19	428	91

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3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

Row	item_purchased	avg_review_rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Handbag	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

Row	shipping_type	avg_purchase_amount
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

Row	subscription...	Total_customer	avg_spend	total_revenue
1	true	1053	59.49	62645
2	false	2847	59.87	170436

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

Row	item_purchased	discount_rate
1	Hat	50.0
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

Row	customer_segment	Number_of_customers
1	Loyal	3116
2	Returning	701
3	New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

Row	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Pants	171
5	2	Clothing	Blouse	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

Row	subscription_status	repeat_customers
1	true	958
2	false	2518

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

Row	age_group	Total_revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

11. **Gender Distribution Overview:** What is the overall distribution of customers by gender?

Row	gender	Total_number
1	Male	2652
2	Female	1248

12. **Subscription by Gender:** How does subscription participation vary between male and female customers?

Row	subscription...	gender	Total_number
1	false	Male	1599
2	false	Female	1248
3	true	Male	1053

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

6. Business Findings & Recommendations

Key Findings

- **Male customers** contributed a larger share of revenue.
- **Subscribers** spend more on average than non-subscribers.
- **Discounts** successfully drive sales but must be balanced with margins.
- **Top-rated products** often align with best-selling categories.
- **Female participation** in purchases and subscriptions is notably low.

Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.
- **Engage Women:** Run campaigns to attract more female customers and add new subscription features designed for them.